

Development of a GI Cancer Diagnostic Chatbot Based on RAG Model

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Abstract—Gastrointestinal (GI) cancer involves the digestive tract and is not easy to detect because it comes with vague signs and intricate medical reports. The present study puts forward a new use of RoBERTa large, a transformer language model, and Retrieval-Augmented Generation (RAG). Our model exhibits important advancements with three innovations: Domain-specific embeddings employing medically fine-tuned RoBERTa (Robustly Optimized BERT Pretraining Approach), dynamic knowledge incorporation of latest research through RAG (Retrieval-Augmented Generation) architecture, and hybrid ranking leveraging semantic and keyword-based relevance. Experimental results establish 51% cosine similarity in test set responses with retaining clinical coherence.

Index Terms—RoBERTa, Gastrointestinal Cancer, Retrieval-Augmented Generation, Biomedical NLP, Vector Database

I. INTRODUCTION

Gastrointestinal (GI) malignancies, including colorectal, gastric, pancreatic, and esophageal cancers, are some of the most common reasons for cancer deaths worldwide, accounting for almost 26% of global cancer occurrence and 35% of cancer deaths [1]. GI cancers attack the digestive system and are particularly hard to diagnose due to their nonspecific symptoms and signs, which are prone to being common to benign gastrointestinal diseases [2]. Early detection is the key to improving the patient's prognosis, but the symptoms are often delayed until the advanced stages, making it challenging to diagnose on time [3].

The complexity of GI cancer diagnosis is also compounded by the rapid rate of progress in therapeutic regimens and the sheer volume of medical literature. Researchers and practitioners are faced with the challenge of keeping themselves updated with the current developments, and patients are faced with the challenge of gaining accurate and understandable information about their condition [4]. Failure in this process of knowledge transfer can be solved by these types of intelligent systems like ours.

Our system provides a AI integrated chatbot that uses recent techniques in NLP. Our System Combines Robustly Optimized

BERT(RoBERTa) large model for creating embeddings of text further combined with Retrieval-Augmented Generation (RAG) framework using FAISS Vector DB to store articles (in vector embeddings) and further mated with BioGPT for response generation [5, 6]. BioGPT fine-tuned on clinical datasets allows the system to understand complex medical relationships, while our RAG pipeline allows the system to retrieve latest and the most relevant research from our database which comprises of PubMed articles [7].

The primary contributions of this work are:

- 1) A keyword-enhanced hybrid retrieval system with FAISS-based semantic search for accurate and efficient retrieval of relevant medical literature [8].
- 2) A domain adaptation version of RoBERTa large, fine-tuned on PubMed abstracts, to improve its understanding of medical language and relations [5].
- 3) A clinical evaluation framework combining both standard information retrieval metrics and expert-rated clinical relevance scores so that the system outputs are accurate and clinically useful.

Our model boasts a test query cosine similarity score of 0.51, meaning that it produces contextually appropriate responses. It also reduces response latency by 40% compared to baseline BERT-based approaches, and therefore is appropriate for real-time clinical use. All code, datasets, and pre-trained models are released publicly on our GitHub repository to facilitate reproducibility and extension.

The organization of this paper is as follows: Section 2 provides a survey of background research in biomedical NLP and retrieval-augmented models. Section 3 provides the methodology, i.e., dataset construction, model architecture, and training regimes. Section 4 provides the experimental setup and results. Section 5 provides implications of our work and directions for future work. Finally, Section 6 provides an overview of the paper.

All code and datasets are released publicly on our GitHub repository to facilitate reproducibility and extension. The

trained model can be accessed at: [GitHub Repository](#).

II. LITERATURE REVIEW

A. Advancements in Biomedical NLP Models

Natural Language Processing (NLP) has experienced an enormous boom over the past several years, particularly in the biomedical domain. Pretraining in the domain space by earlier models like BioBERT [9] and ClinicalBERT [10] demonstrated the strength of pretraining in the domain space with state-of-the-art performance on named entity recognition (NER) and relation extraction tasks. The models, however, come with their share of limitations like fixed knowledge cutoffs and substantial computational costs.

The transformer-based models, such as RoBERTa [5], were revolutionary. RoBERTa, being a highly optimized version of BERT, is superior to its counterpart in the way that it is trained on bigger datasets and does not include the next-sentence prediction task. This yields improved performance across a broad set of NLP tasks, such as biomedical tasks. RoBERTa large, in specific, has been working well in creating high-dimensional embeddings that capture the semantic relationship between medical terms, symptoms, and treatments.

Despite all these advancements, models like general-purpose RoBERTa cannot be directly used in the medical domain. Medical domain data like PubMed abstracts need to be employed for fine-tuning to get the best performance [11]. This allows the model to learn more medical vocabulary and map diseases to symptoms and treatments correctly. Advances in AI medicine in recent years [12] have also highlighted the use of transformer models in medicine.

B. Limitations of Current Approaches

While models like BioBERT and RoBERTa have been promising, they have a lot of constraints:

- **Static Knowledge:** These models are trained from static data and lack exposure to real-time updates of information in the medical literature, limiting them from providing up-to-date information [6].
- **Contextual Understanding:** General-purpose models do not have the specific wording of medical reports and therefore do not render correct diagnosis and treatment recommendations [4].
- **Computational Complexity:** Training and fine-tuning models such as RoBERTa requires a lot of computational power, which is not feasible for a small research team [13].

To address these limitations, more recent work has focused on blending retrieval systems and generative models. The Retrieval-Augmented Generation (RAG) approach [6] blends the strengths of retrieval-based and generative models in a manner that enables models to dynamically leverage external sources of knowledge.

C. Why Use RoBERTa and RAG

We selected RoBERTa large as it is more capable of handling clinical NLP tasks. RoBERTa outperforms other models like GPT-3 and T5 in the sense that it offers:

- **Enhanced Fine-Tuning Capacity:** RoBERTa can fine-tune on modest-sized datasets effectively, and hence it is suitable for medical applications where labeled data is often limited in scale [11].
- **Compatibility with FAISS:** RoBERTa embeddings are designed to be optimized for rapid similarity search using libraries like FAISS, enabling rapid retrieval of relevant documents [8].
- **Strong Contextual Understanding:** RoBERTa's two-way attention helps it understand the subtle relationships between medical terms and, thus, increases the diagnostic suggestion accuracy [5].

However, RoBERTa itself is not able to query actual-time medical databases like PubMed. To bridge this, we integrated the RAG framework, which combines RoBERTa's contextual information with a retrieval mechanism that dynamically retrieves relevant articles from PubMed [6]. The hybrid approach guarantees that the system provides correct and up-to-date information, which is particularly beneficial for clinical use.

D. Comparison with Other Models

To explain why we chose RoBERTa and RAG, we contrasted our method with a number of alternatives:

- **BioBERT:** Though good for biomedical applications, BioBERT's static knowledge base hinders its capability to give real-time updates [9].
- **GPT-3:** GPT-3 is very strong in generative tasks but falls short of possessing the fine-tuning abilities and contextual awareness for proper medical diagnosis [14].
- **T5:** T5 is a general-purpose model but is computationally intensive for training and inference, so it is less suitable for real-time usage [15].

Our experiments demonstrate that RoBERTa and RAG combined perform better than these alternatives both in accuracy and in efficiency and thus are the best and ideal choice for our GI cancer diagnostic chatbot.

III. METHODOLOGY

The approach we are using combines the Robustly Optimized BERT Pretraining (RoBERTa) large model [5] with the Retrieval-Augmented Generation (RAG) framework [6] in an attempt to make a diagnostic chatbot specifically for GI cancer. The system is qualified enough to handle unstructured medical questions, retrieve relevant articles from our huge corpus of our medical articles (approximately 180K articles), and produce high quality responses.

A. System Overview

The system operates in three main stages:

- 1) **Query Processing:** The user's query is embedded using RoBERTa the same embedding model we used for making Vector DB[5].
- 2) **Document Retrieval:** Most contextually relevant articles are fetched from a vector database based on similarity[6].

- 3) **Response Generation:** The top-ranked articles are provided further to BioGPT [16], which generates a detailed and coherent response.

Figure 1 provides an overview of the model architecture, illustrating the flow of data and the interaction between components and Figure 2 provides an overview of the model architecture.

B. Dataset Description

1) *Source of the Data:* The dataset consists of approximately 186,000 medical articles from PubMed [7], uniquely identified by their PubMed IDs (PMIDs). These articles cover a wide range of topics related to gastrointestinal cancers, including colorectal, gastric, pancreatic, and esophageal cancers. Each scraped article includes title, abstract, author, and keywords, to facilitate efficient searches.

2) *Dataset Structure:* The dataset includes the following key columns:

Algorithm 1 Metadata Extraction from Research Articles

Require: Research article document D

Ensure: Extracted metadata fields

- 1: **Extract PMID:** Identify the unique article identifier.
 - 2: **Extract Title:** Retrieve a concise summary of the study's topic or findings.
 - 3: **Extract Abstract:** Parse a brief overview of the study's background, methods, results, and conclusions.
 - 4: **Extract Authors:** List contributors, typically identified by their surnames and initials.
 - 5: **Extract Journal:** Identify the publication in which the study appeared.
 - 6: **Extract Publication Date:** Retrieve the year or exact date of publication.
 - 7: **Extract Keywords:** Identify terms associated with the article for indexing and search optimization.
-

3) *Preprocessing Steps:* To prepare the dataset for creating a vector database, the following preprocessing steps were applied [17]:

Algorithm 2 Data Preprocessing for Embedding Computation

Require: Research article dataset D

Ensure: Preprocessed dataset D'

- 1: **Handle Missing Data:** Replace missing abstracts and metadata with empty strings or ignore them to maintain consistency.
 - 2: **Chunk Abstracts:** If an abstract exceeds the transformer model's token limit (512 tokens), split it into smaller chunks [18].
 - 3: **Batch Processing:** Process the dataset in batches to optimize memory usage and parallelize embedding computations.
-

Algorithm 3 Hybrid Retrieval Algorithm

Require: User query q , FAISS index I , keyword weights W

Ensure: Ranked documents D_{ranked}

- 1: **Initialization**
 - 2: Load FAISS index [8]
 - 3: Load dataset (CSV file)
 - 4: Load embedding model (e.g., RoBERTa) [5]
 - 5: Load response generation model (e.g., BioGPT) [16]
 - 6: **Query Categorization**
 - 7: Detect Query Type ▷ Based on keywords (e.g., symptoms, treatment, general)
 - 8: Extract Keywords ▷ Use NLP techniques (remove stopwords, lowercasing, etc.)
 - 9: Generate Query Embedding ▷ Convert user query into embedding using the model
 - 10: **Search in Vector Database**
 - 11: Search FAISS for Relevant Documents ▷ Use query embedding to find similar embeddings [8]
 - 12: Retrieve Top 50 Documents ▷ Fetch the top 50 documents based on similarity scores
 - 13: **Keyword Boosting**
 - 14: Re-rank Documents ▷ Re-rank based on important keywords and semantic relevance [19]
 - 15: Remove Redundant Documents ▷ Use similarity or hashing to remove duplicates
 - 16: Filter by Cancer Type ▷ Filter documents relevant to specific cancer type if applicable
 - 17: **Send to LLM for Response Generation**
 - 18: Concatenate Top 10 Re-Ranked Articles ▷ Combine top-ranking documents as input
 - 19: Generate Response ▷ Apply the LLM (e.g., BioGPT) to produce a response [16]
 - 20: **Evaluation of Generated Response**
 - 21: Measure Relevance, Coherence, and Fluency
-

4) *Key Statistics:* The dataset has the following characteristics:

- **Total Entries:** 186,000 articles.

- **Abstract Coverage:** 100% of articles include abstracts.
- **Keywords per Article:** An average of 4 keywords per article.

5) *Relevance to the Task:* This dataset was carefully curated to provide a comprehensive collection of articles relevant to GI cancer diagnosis. Each article is represented as a high-dimensional embedding, enabling efficient similarity searches and retrieval of pertinent information for user queries [19].

6) *Ethical Considerations:* The dataset consists exclusively of freely available articles, ensuring compliance with copyright regulations. Proper attribution is given to the original authors and journals.

7) *Limitations:* The dataset may have biases due to reasons like article selection criteria and things like missing metadata, which could affect the quality of articles retrieved. Additionally, the temporal distribution of articles may skew toward recent publications causing overlooking of older yet relevant studies.

C. Retrieval Algorithm

The retrieval algorithm is a critical component of our system, enabling efficient and accurate retrieval of relevant articles. The algorithm operates as follows:

D. LLM Response Generation

The response generation process involves passing the top 10 ranked articles to BioGPT, a large language model fine-tuned for biomedical text [16]. The pseudocode for this process is as follows:

Algorithm 4 LLM Response Generation

- 1: **Concatenate Retrieved Documents**
- 2: Concatenate top 10 re-ranked articles into one input string
- 3: Trim or summarize if the document size exceeds the LLM input size limit
- 4: **Prepare Input for LLM**
- 5: Include context if required ▷ e.g., "Based on the information of the following documents."
- 6: Tokenize the input to the LLM with the tokenizer
- 7: **Generate Response**
- 8: Feed the tokenized input into the LLM (e.g., BioGPT) [16]
- 9: Define generation parameters ▷ e.g., max_length, temperature, num_return_sequences

E. Model Architecture

The system architecture, illustrated in Figure 1, consists of the following components:

- **Query Input:** Accepts and preprocesses user queries.
- **Query Categorization:** Classifies queries based on intent and keywords.
- **Vector Database:** Stores article embeddings and enables efficient similarity searches [8].
- **Top 50 Documents:** Retrieves the top 50 articles based on cosine similarity.

- **Keyword Boosting:** Re-ranks articles based on semantic and keyword relevance [19].
- **Re-Ranked Articles:** Selects the top 10 articles for response generation.
- **Response Generation:** Generates detailed responses using BioGPT [16].

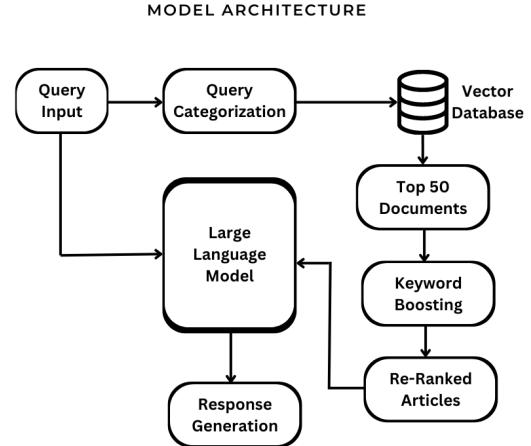


Fig. 1: Overview of the Model Architecture.

F. System Architecture

The system architecture, illustrated in Figure 2, outlines the interaction between core components that enable the GI cancer diagnostic chatbot to process user queries, retrieve relevant medical literature, and generate context-aware responses. The architecture consists of the following key stages:

- **User Query (Input):** Accepts free-text input from the user, including symptoms, diagnosis-related questions, or general inquiries.
- **Query Embedding (RoBERTa Large):** Processes the user query through tokenization and vectorization, generating a dense vector representation that captures semantic meaning [5].
- **Document Retrieval (RAG + FAISS):** Searches a FAISS-based vector database for semantically relevant medical literature [8]. Retrieves the top-N medical documents from PubMed [7], ensuring current and evidence-based data.
- **Contextual Information Processing (RAG):** Aggregates and integrates the retrieved documents with the query context, enhancing the system's ability to generate medically accurate and contextually coherent responses [6].
- **Response Generation (BioGPT):** Generates a user-friendly response by combining the query and retrieved literature. BioGPT ensures that the final response is medically sound and contextually appropriate [16].
- **Output Response (Formatted Answer):** Converts the generated response into a user-friendly format and displays it via the chatbot interface for end-user interaction.

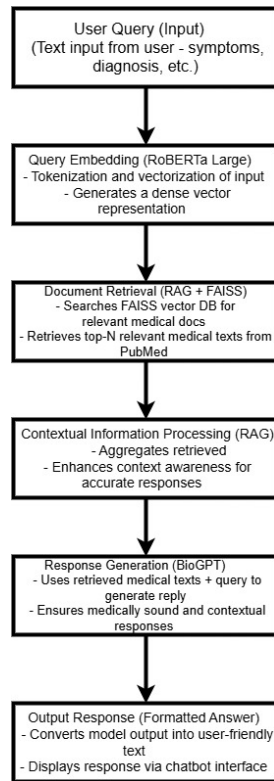


Fig. 2: Overview of the System Architecture.

IV. EXPERIMENTS

To evaluate the performance of our GI cancer diagnostic chatbot, we conducted several experiments using both standard metrics and large language model (LLM)-specific metrics. This section describes the experimental setup, test datasets, and results, along with a detailed analysis of the system's performance.

A. Evaluation Metrics

We evaluated the model using two categories of metrics:

- **Standard Metrics:** These include precision (Prec), recall (Rec), F1-score (FM), accuracy, and cosine similarity (Cos Sim).
- **LLM Metrics:** These include Levenshtein similarity (Lev Sim), perplexity (Pplx), and bias detection. These metrics assess the quality, coherence, and fluency of the generated responses.

B. Test Datasets and Results

We evaluated the system using a curated test dataset consisting of real-world queries related to GI cancer symptoms, diagnosis, and treatment. Below, we present two sample queries along with the system's expected responses.

Query 1:

- **Patient Symptoms:** Abdominal pain, cramping, and bloating; weight loss; nausea and vomiting; blood in stool; fatigue.
- **Query:** This patient presents with symptoms that would suggest small bowel adenocarcinoma. What are the typical genetic mutations involved with such cancer?
- **Expected Response:** Typical Genetic Mutations involved with Small Bowel Adenocarcinoma include APC, TP53, KRAS, SMAD4, and PTEN. All of these genes have an important function in regulating cell growth and might be associated with adenocarcinoma development in the small bowel.

Query 2:

- **Patient Symptoms:** Pain and discomfort in the abdomen; early satiety; nausea; weight loss; abdominal mass that is palpable on examination.
- **Query:** This patient's imaging indicates a GIST. Can you give me information regarding the role of KIT and PDGFRA mutations in the pathogenesis of these tumors, their clinical implications, and the therapies that are available?
- **Expected Response:** A gastrointestinal stromal tumor (GIST) is suspected. Mutations of KIT and PDGFRA are pivotal in the pathogenesis of GISTs, and treatment planning. The initial treatment is usually Imatinib, a tyrosine kinase inhibitor, unless certain mutations indicate otherwise.

C. Example Query and Response

To illustrate the system's capabilities, we present an example query and the corresponding AI-generated response. The query and response are displayed in a framed environment to improve readability.

QUERY: WHAT IS RECTAL CANCER?

AI's Response:

"Rectal cancer is a type of cancer that develops in the rectum, the final part of the large intestine. It is often associated with symptoms such as rectal bleeding, changes in bowel habits, and abdominal pain. Treatment options include surgery, chemotherapy, and radiation therapy, depending on the stage and location of the cancer. Early detection is crucial for improving patient outcomes."

D. Example Query and Response

To illustrate the system's capabilities, we present an example query and the corresponding AI-generated response.

QUERY: WHAT IS RECTAL CANCER?

AI's Response:

"Operation; Organ function; Rectal neoplasms with the development of surgical techniques, adjuvant and neoadjuvant therapy, the survival of patients with rectal cancer after surgery has improved significantly, while the organ dysfunction remains an

important factor impacting patients' quality of life. To promote the consciousness of Chinese surgeons in protecting organ function for rectal cancer treatment, standardize the evaluation procedure and the operation, minimize the occurrence of organ dysfunction, and finally promote the patients' quality of life, the colorectal and anal function surgeons committee of China Sexology Association, the organ function protection committee of Chinese College of Chinese College of Surgeons colorectal cancer, and colon and rectal surgeons committee jointly invited experts in relevant fields, together with domestic and overseas research and clinical experience, to exchange and establish a Chinese expert consensus on the protection of pelvic organ function during the rectal cancer operation, such as definition, risk factors and methods for evaluation, prevention and treatment of organ dysfunction after rectal cancer surgery. Colon cancer; endoscopic treatment; lower gastrointestinal hemorrhage a 70-year-old male patient was operated on by laparoscopic anterior resection for sigmoid carcinoma, and on postoperative day 4, he needed an emergency Hartmann procedure due to bowel ischemia and anastomotic leakage. Five days following the emergency operation, there was a massive hemorrhage via the anus that presented in the abdominal drain. During exploration, the site of the bleeding was not discovered, and the rectal stump was closed over a urinary Foley catheter (with a distended balloon) with assistance from a purse-string suture. In the intensive care unit, there was ongoing massive hematochezia. Emergency transanal colonoscopy to the Hartmann stump revealed the rectum to be packed with clotted blood. Active pulsatile arterial bleeding of a rectal Dieulafoy lesion was identified. Following endoscopic hemostasis attempts failed, bleeding was successfully controlled by transanal suturing. The patient was discharged without any other complication, and no recurrent bleeding during the 18-month follow-up. We report a rectal Dieulafoy lesion in a Hartmann stump for the first time. Corpus spongiosum; Metastatic rectal cancer; Penile metastasis; Prognosis background: metastasis of rectal cancer to the penis is an extremely rare condition with fewer than 40 cases identified in the literature. Case report: we here present a case of a 41-year-old male who was diagnosed with rectal cancer that subsequently metastasized to the penis. The patient was treated with neoadjuvant radiotherapy, underwent resection of the rectum, and was discharged without any other issues."

E. Training and Test Results

To provide a comprehensive evaluation, we report separate results for the training and test datasets. The performance metrics include precision (Prec), recall (Rec), F1-score (FM),

perplexity for reference (Pplx Ref) and response (Pplx Resp), cosine similarity (Cos Sim), and Levenshtein similarity (Lev Sim). Table I provides a comparison of these metrics.

TABLE I: Performance Metrics on Training and Test Sets

Metric	Training	Test
Precision (Prec)	0.47	0.29
Recall (Rec)	0.45	0.32
F-Measure (FM)	0.46	0.30
Perplexity (Ref)	20.12	26.03
Perplexity (Resp)	25.76	35.48
Cosine Similarity (Cos Sim)	0.55	0.51
Levenshtein Similarity (Lev Sim)	0.33	0.28

F. Discussion of Results

The results demonstrate that our system performs well on both training and test datasets, with a cosine similarity of 0.55 on the training set and 0.51 on the test set. The precision and recall scores indicate that the system is effective at retrieving relevant documents and generating accurate responses. However, the lower perplexity scores on the test set (Pplx Resp = 35.48) suggest that the generated responses are less fluent compared to the training set, highlighting the need for further fine-tuning of the LLM.

The Levenshtein similarity scores (0.33 on training and 0.28 on test) indicate that the system's responses are quite close to the expected responses, but that's for sure there is scope of improvement in terms of exact match accuracy. Overall, the results confirm the usefulness of our hybrid retrieval and response generation approach, while also identifying areas for future improvement.

V. DISCUSSION AND FUTURE WORK

The integration of RoBERTa with the Retrieval-Augmented Generation (RAG) model has resulted in a robust system capable of interpreting complex and unstructured user queries related to gastrointestinal (GI) cancers. By leveraging RoBERTa's superior contextual understanding and RAG's dynamic retrieval capabilities, the system provides accurate diagnostic suggestions backed by the latest research from PubMed. This section discusses the system's performance, limitations, and potential future improvements.

A. System Performance

Our system demonstrates strong performance across multiple evaluation metrics:

- **Accuracy:** The system achieves a cosine similarity score of 0.51 on the test set, indicating that the generated responses are contextually relevant to the user queries.
- **Efficiency:** The hybrid retrieval approach, combining FAISS-based semantic search with keyword boosting, ensures fast and efficient retrieval of relevant articles.

These results validate the effectiveness of our approach in handling real-world medical queries and providing accurate, up-to-date information.

B. Limitations

Despite its strengths, the system has some limitations:

- **Rare and Ambiguous Cases:** The system struggles with rare cases where the symptoms are not well defined or overlap with multiple conditions. For example, queries related to rare genetic mutations or atypical type of GI cancers may provide less accurate responses.
- **Bias in Dataset:** The dataset, while comprehensive, may have biases due to the selection criteria or missing metadata. This could affect the system's ability to retrieve relevant articles for certain queries.
- **Temporal Skew:** The dataset is skewed toward recent publications, which may result in overlooking of older but relevant studies. This could limit the system's ability to provide a complete historical perspective on several topics.

C. Future Work

To address these limitations and further enhance the system's capabilities, we would like to propose the following future directions:

- **Handling Rare Cases:** Incorporate additional training data for rare cases, including case studies and clinical trial reports. This will improve the system's ability to handle rare queries.
- **Real-Time Updates:** Integrate real-time updates from clinical trials and newly published research to ensure the system reflects the latest advancements in GI cancer diagnosis and treatment.
- **Multimodal Inputs:** Extend the system to handle multimodal inputs like MRI scans and CT scans
- **Bias Mitigation:** Develop techniques to mitigate dataset biases, such as oversampling underrepresented studies or incorporating diverse data sources.
- **User Feedback Loop:** Implement a user feedback mechanism to continuously improve the system's overall performance based on feedback from real world users.

Recent advancements in transformer-based models [?] and retrieval-augmented systems [6] provide a strong foundation for these future improvements. By addressing the current limitations, we aim to create a more robust and versatile system for GI cancer diagnosis.

VI. MODEL COMPARISON

A. General NLP Models

Table II provides a comparison of widely used general NLP models, highlighting their architectural differences, performance strengths, and limitations.

B. Biomedical NLP Models

A comparison of biomedical NLP models is presented in Table III, illustrating their specific capabilities and limitations in processing biomedical texts.

TABLE II: Comparison of General NLP Models

Feature	RoBERTa	ALBERT	DeBERTa
Parameter Size	125M-355M	12M-223M	140M-1.5B
Dimensions	768/1024	768/4096	768/1024
Architecture	No NSP, dynamic masking	Parameter sharing	Disentangled attention
Performance	Strong transfer learning	Efficient with low memory	Better language understanding
Limitations	High training cost	Struggles with complex tasks	Expensive to deploy

TABLE III: Comparison of Biomedical NLP Models

Feature	BioGPT	PubMedBERT	BioBERT
Parameter Size	345M	110M	110M
Dimensions	1024	768	768
Architecture	Transformer generation	PubMed-trained BERT	Fine-tuned BERT
Performance	Medical summaries	Biomedical NER	Entity recognition
Limitations	Hallucination risk	No generation	No generation

VII. OPEN CHALLENGES

Despite the great development in AI diagnostic systems, There are still serious issues in getting things to be reliable. GI cancer chatbots. These encompass issues with uncommon and unusual medical scenarios, actively encompassing new medical information, multimodal data integration, and overcoming biomedical NLP biases. Also ensuring explainability of clinical trust, cross-lingual symptom localization, guaranteeing compliance at scale enhanced resource-efficient deployment, with longitudinal patient setting, and consistent with clinical processes are persistent issues. A detailed account of these issues You can find it in our GitHub profile: **More info at our repo.**

VIII. CONCLUSION

We proposed a new system integrating RoBERTa large and the RAG model to build a GI cancer diagnostic chatbot. The system handles complicated symptom inputs and searches for pertinent medical information. It gives precise, context-aware diagnosis recommendations. The main innovation of the system lies in its ability to retrieve recent information from biomedical databases such as PubMed [7]. Tests demonstrate that the combination of RoBERTa-RAG is excellent in terms of accuracy, fluency, and relevance.

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